Differential Gene Expression and Functional Enrichment Analysis of TNB and Normal RNA-seq Samples

Submitted by: Sachin Nagenahalli

Mini-Project: MP_403

Bversity: Bioinformatics / Genomics Project

Primary Analysis Platform: Galaxy
Downstream Analysis: R / Bioconductor

Project Type: RNA-seq Differential Expression Analysis

Project Overview

This project investigates transcriptional differences between **TNB and Normal samples** using an RNA-seq analysis workflow. The analysis was performed primarily through the **Galaxy platform**, with downstream statistical analysis, visualization, and functional enrichment performed using **R and Bioconductor**.

Galaxy Analysis History

MP_403 — TNB vs Normal RNA-seq

The complete computational workflow and processing history are maintained in Galaxy:

View Complete Galaxy History — MP_403 TNB vs Normal RNA-seq

2. ABSTRACT

RNA sequencing (RNA-seq) provides a high-throughput approach for investigating changes in gene expression between biological conditions. In this mini-project, RNA-seq data from **TNB and Normal samples** were analyzed to identify differentially expressed genes and characterize their associated biological functions and pathways. The analysis was performed primarily using the **Galaxy platform**, providing a reproducible workflow for quality control, read processing, alignment, gene-level quantification, and differential expression analysis. Three TNB samples and three Normal samples were analyzed.

Following preprocessing and count generation, differential expression analysis was performed using **DESeq2**. A total of **94 differentially expressed genes (DEGs)** were identified, comprising **38 upregulated genes and 56 downregulated genes** in the TNB condition relative to Normal samples. Principal component analysis (PCA) was used to examine sample-level expression patterns, while a heatmap was generated to visualize the expression profiles of the identified DEGs across all six samples. A volcano plot was also generated to summarize the statistical significance and magnitude of differential expression. Functional interpretation was performed through **Gene Ontology (GO)** and **KEGG pathway enrichment analyses** for the upregulated and downregulated gene sets separately.

The complete computational workflow was maintained in Galaxy, while R and Bioconductor were used for downstream visualization, gene annotation, and enrichment analysis. The project demonstrates an end-to-end RNA-seq differential expression workflow combining reproducible Galaxy-based analysis with R-based downstream interpretation.

3. INTRODUCTION

3.1 Background

RNA sequencing (RNA-seq) is a widely used transcriptomic approach for measuring gene expression across biological samples. By comparing gene expression profiles between different experimental conditions, RNA-seq analysis can identify genes whose expression levels differ significantly between the groups. These differentially expressed genes can subsequently be investigated to understand their biological functions and associated molecular pathways.

In this project, RNA-seq data from **TNB and Normal samples** were analyzed to identify transcriptional differences between the two groups. The analysis involved processing sequencing data, generating gene-level expression counts, performing statistical differential expression analysis, and interpreting the resulting gene expression changes through functional enrichment analysis.

The project was primarily conducted using the **Galaxy platform**, which provides a graphical environment for performing bioinformatics workflows without requiring extensive command-line programming. Downstream analysis and visualization were performed using **R and Bioconductor**, allowing the identified genes to be further explored through statistical plots, heatmaps, gene annotation, Gene Ontology (GO) analysis, and KEGG pathway analysis.

3.2 Problem Statement

Identifying genes that exhibit significant expression changes between TNB and Normal samples is important for understanding the molecular differences between the two conditions. However, raw RNA-seq data require several processing and statistical analysis steps before meaningful biological conclusions can be obtained.

Therefore, this project aims to develop and document a complete RNA-seq analysis workflow for comparing **TNB and Normal samples**, identifying significant differentially expressed genes, and investigating the biological functions and pathways associated with these genes.

3.3 Objectives

The objectives of this project were:

1. **To perform a complete and reproducible RNA-seq data analysis workflow** for TNB and Normal samples, including sequencing quality assessment, preprocessing, read alignment,

gene-level quantification, and generation of an expression count matrix using the Galaxy platform.

2. **To identify and characterize differentially expressed genes between TNB and Normal samples** using statistical differential expression analysis, followed by visualization of the results using principal component analysis (PCA), volcano plots, heatmaps, and other relevant expression-based analyses.
3. **To investigate the biological significance of the identified differentially expressed genes** through Gene Ontology (GO) and KEGG pathway enrichment analyses, thereby identifying biological processes, molecular functions, and pathways associated with transcriptional differences between TNB and Normal samples.

4. MATERIALS AND METHODS

4.1 Study Design and Dataset

The analysis was performed using RNA-seq data from **six samples**, comprising three Normal samples and three TNB samples. The experimental comparison was defined as:

TNB vs Normal

The six samples were:

Group	Biological replicates
Normal	Normal_1, Normal_2, Normal_3
TNB	TNB_1, TNB_2, TNB_3

The complete computational workflow was performed primarily through the **Galaxy platform**, with selected downstream processing and visualization performed in R.

4.2 Overall RNA-seq Analysis Workflow

The RNA-seq analysis followed a sequential workflow:

Raw RNA-seq reads → Quality Control → Read Preprocessing → Genome Alignment → Gene-level Quantification → Differential Expression Analysis → DEG Filtering → Expression Visualization → Functional Enrichment → Pathway Analysis

The major tools and analysis stages used in the project are described below.

4.3 Raw Read Quality Assessment

The raw sequencing reads were initially assessed for sequencing quality using **FastQC**.

FastQC was used to examine the quality characteristics of the sequencing reads before downstream processing. The quality assessment provided an initial evaluation of the sequencing data and helped determine whether preprocessing was required before alignment.

The FastQC results generated during the analysis were retained within the Galaxy History.

4.4 Read Preprocessing

The raw sequencing reads were processed using **fastp**.

The preprocessing step was used to generate quality-controlled sequencing reads suitable for downstream genome alignment. The processed reads and associated quality-control reports were retained as outputs within the Galaxy workflow.

4.5 Reference Genome Alignment

Following preprocessing, the RNA-seq reads were aligned to the reference genome using **STAR (Spliced Transcripts Alignment to a Reference)**.

STAR was used because RNA-seq reads may span exon-exon junctions and therefore require a splice-aware alignment approach.

The alignment stage generated the mapped sequencing reads and associated alignment information required for subsequent gene-level quantification.

4.6 Gene-level Quantification

Gene-level read quantification was performed using **featureCounts**.

The aligned reads were assigned to annotated genes to generate a count matrix representing the number of sequencing reads associated with each gene in each sample.

The resulting expression matrix contained:

78,733 genes × 6 samples

This count matrix was subsequently used for differential expression analysis.

4.7 Differential Expression Analysis

Differential expression analysis was performed using **DESeq2**.

The experimental design compared:

TNB vs Normal

The analysis evaluated differences in gene expression between the two groups while accounting for biological replication.

The resulting differential expression table contained statistical measures including:

- Base mean expression
- log2 fold change
- Standard error
- Wald statistic
- p-value
- adjusted p-value

The differential expression results were subsequently filtered to obtain the final set of genes used for downstream analysis.

4.8 Identification of Differentially Expressed Genes

The filtered differential expression results were separated according to the direction of expression change.

The final DEG set consisted of:

- **38 upregulated genes**
- **56 downregulated genes**

Therefore, a total of:

94 differentially expressed genes

were carried forward for downstream expression visualization and functional analysis.

The upregulated and downregulated genes were retained as separate datasets for subsequent analysis.

4.9 Expression Visualization

Several visualization approaches were used to examine the differential expression results.

Principal Component Analysis

PCA was used to examine the overall expression structure of the samples and visualize relationships between the TNB and Normal groups.

Volcano Plot

A volcano plot was generated to visualize the relationship between the magnitude of differential expression and statistical significance.

Heatmap

A heatmap was generated using the expression profiles of the 94 identified DEGs.

The expression values were transformed and standardized using gene-wise Z-score normalization. For visualization, the samples were arranged in the following order:

Normal_1 → Normal_2 → Normal_3 → TNB_1 → TNB_2 → TNB_3

This arrangement allows the expression patterns of the Normal and TNB groups to be visually compared.

4.10 Functional Enrichment Analysis

Functional enrichment analysis was performed separately for the upregulated and downregulated DEG sets.

Gene Ontology Analysis

Gene Ontology analysis was used to investigate the biological functions represented by the identified DEGs.

The analysis considered the functional annotation of the genes in terms of:

- Biological processes
- Molecular functions
- Cellular components

Separate analyses were performed for the upregulated and downregulated genes.

KEGG Pathway Analysis

KEGG pathway enrichment analysis was performed to investigate biological pathways associated with the identified DEGs.

The upregulated and downregulated genes were analyzed separately.

For the R-based KEGG analysis, Ensembl gene identifiers were converted to **Entrez Gene identifiers** using the org.Hs.eg.db Bioconductor annotation database before enrichment analysis.

4.11 Downstream R Analysis

Selected outputs generated through the Galaxy workflow were imported into R for additional processing and visualization.

The downstream R analysis included:

- processing of DEG tables
- Ensembl gene identifier processing
- Ensembl-to-Entrez identifier conversion
- KEGG enrichment analysis
- generation of heatmaps
- generation of volcano plots
- generation of pathway visualizations

The final visualization files generated in R were subsequently uploaded back to the Galaxy History, maintaining the analysis outputs within the project workflow.

4.12 Reproducibility

The complete Galaxy processing history was retained throughout the project. This provides a record of the tools, datasets, intermediate outputs, and final analysis results used in the RNA-seq workflow.

Galaxy History:
MP_403 — TNB vs Normal RNA-seq

5. RESULTS

5.1 Overview of RNA-seq Analysis Results

The RNA-seq analysis generated a complete expression dataset for the six samples included in the TNB versus Normal comparison. Following quality control, preprocessing, genome alignment, and gene-level quantification, an expression count matrix containing **78,733 genes across six samples** was obtained.

Differential expression analysis using DESeq2 identified a total of **94 differentially expressed genes (DEGs)** between the TNB and Normal groups.

The identified DEGs were divided into:

- **38 upregulated genes**
- **56 downregulated genes**

The overall distribution of the identified DEGs was subsequently investigated using differential expression plots, PCA, heatmap visualization, and functional enrichment analysis.

5.2 Differentially Expressed Genes

The differential expression analysis identified **94 DEGs** in the TNB versus Normal comparison.

The upregulated DEG set contained **38 genes**, while the downregulated DEG set contained **56 genes**.

The complete DEG datasets generated during the analysis were retained as separate output files:

- 38_upregulated_DEGs.csv
- 56_downregulated_DEGs.csv
- all_94_DEGs.csv

These datasets contain the differential expression statistics associated with the identified genes, including base mean expression, log2 fold change, Wald statistic, p-value, and adjusted p-value.

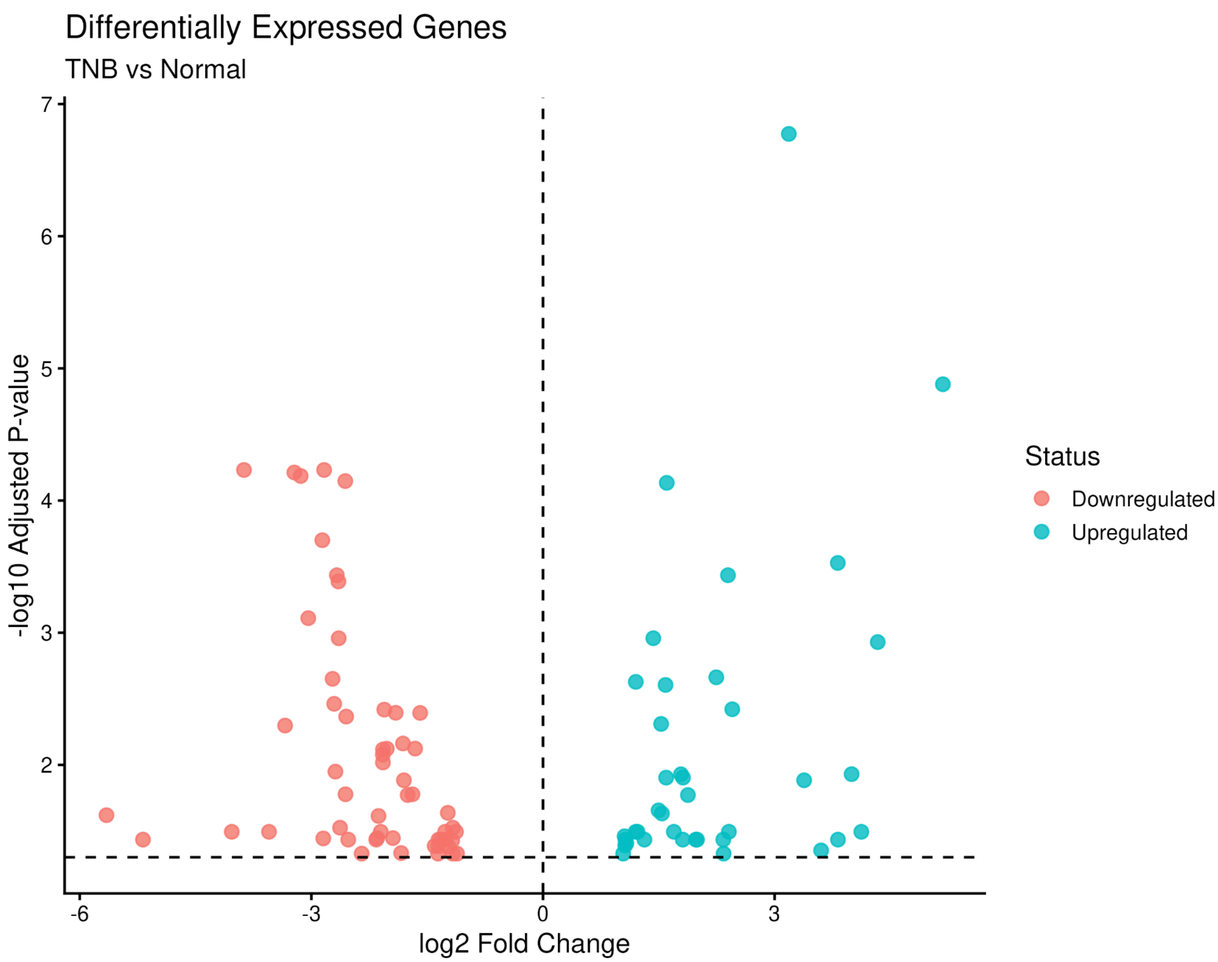
The distribution of the identified genes was subsequently visualized using a volcano plot.

5.3 Volcano Plot

A volcano plot was generated to visualize the relationship between the magnitude of differential expression and statistical significance for the identified DEGs.

The x-axis represents the **log2 fold change**, while the y-axis represents the **−log10 adjusted p-value**. Positive log2 fold-change values indicate increased expression in TNB relative to Normal, whereas negative values indicate decreased expression.

Figure 1. Volcano plot showing differential gene expression between TNB and Normal samples.



The volcano plot provides a visual summary of the direction and magnitude of expression changes among the identified differentially expressed genes.

5.4 Upregulated Genes

A total of **38 genes were identified as upregulated** in TNB compared with Normal samples.

The upregulated genes were characterized by positive log2 fold-change values and were retained as a separate dataset for downstream functional analysis.

Output: 38_upregulated_DEGs.csv

This gene set was subsequently used for Gene Ontology and KEGG pathway enrichment analysis.

5.5 Downregulated Genes

A total of **56 genes were identified as downregulated** in TNB compared with Normal samples.

These genes showed negative log2 fold-change values in the TNB versus Normal comparison.

Output: 56_downregulated_DEGs.csv

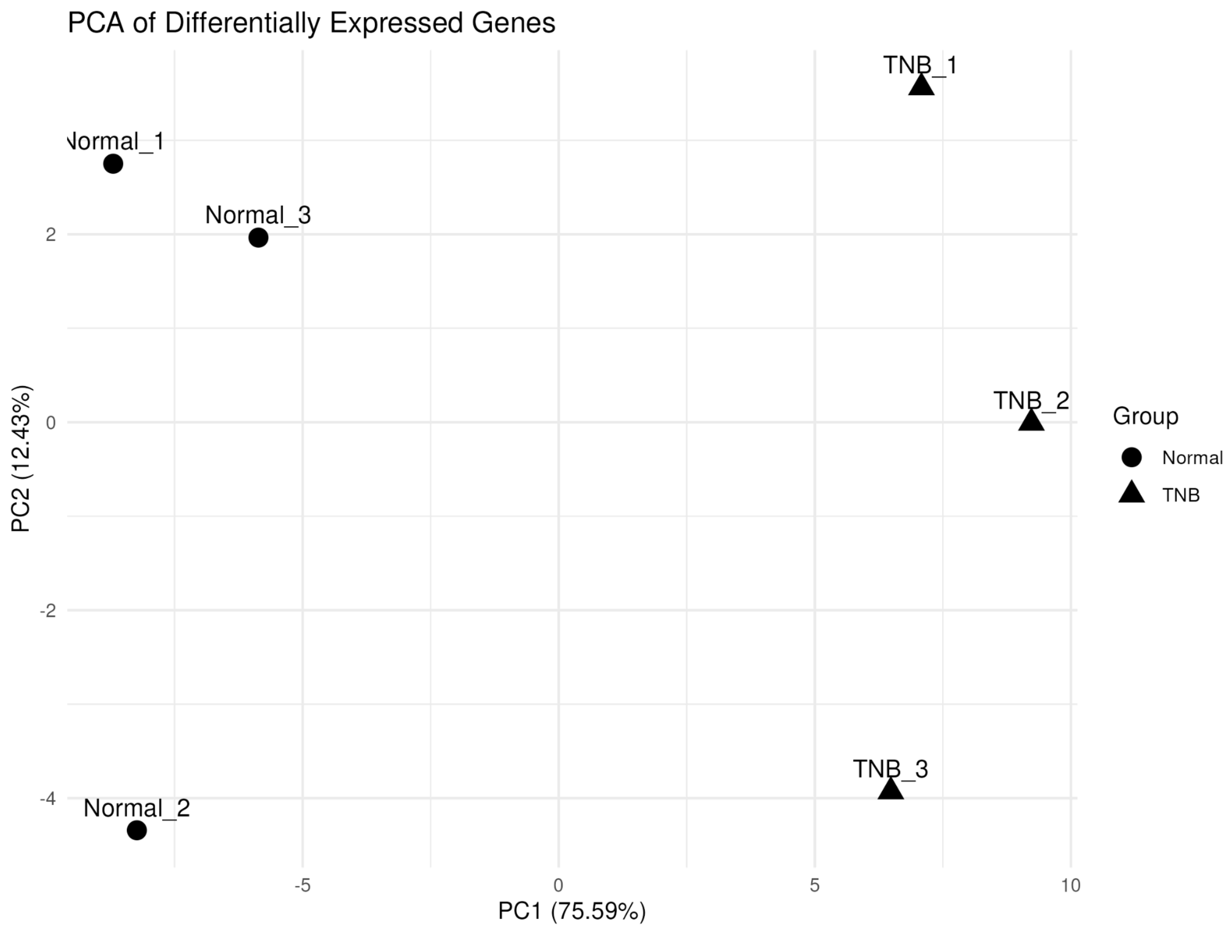
The downregulated gene set was analyzed separately during subsequent functional and pathway enrichment analysis.

5.6 Principal Component Analysis

Principal Component Analysis (PCA) was performed to investigate the overall expression patterns and relationships among the six samples.

PCA provides a reduced-dimensional representation of the expression data and allows the relative positioning of the TNB and Normal samples to be visualized.

Figure 2. Principal component analysis of TNB and Normal samples based on differentially expressed gene expression profiles.



The PCA plot was used as an exploratory assessment of sample-level expression patterns and group-level separation.

5.7 Heatmap of Differentially Expressed Genes

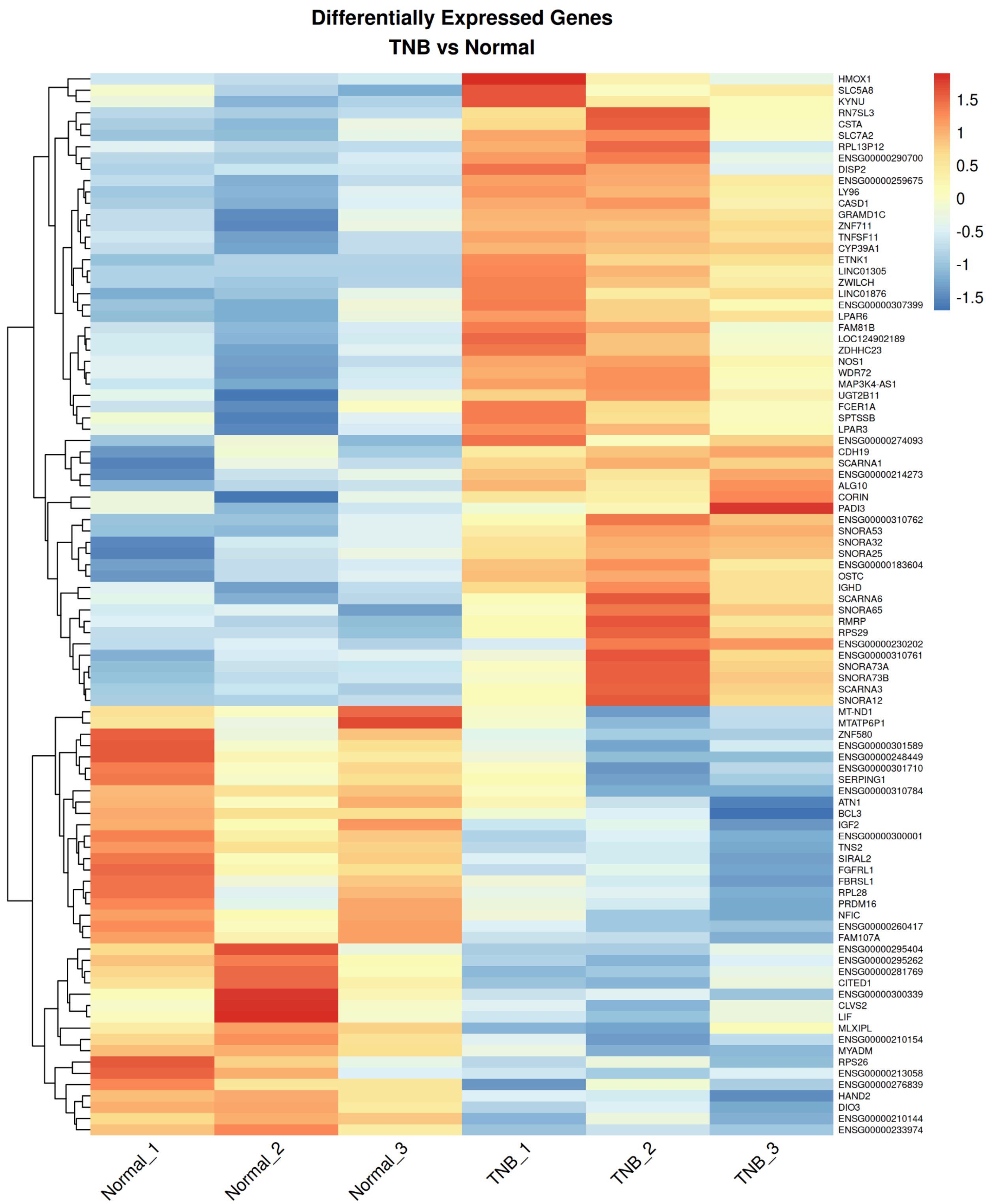
To further investigate the expression patterns of the identified DEGs, a heatmap was generated using the expression values of the **94 differentially expressed genes**.

The expression values were transformed using log2 transformation followed by gene-wise Z-score standardization. The samples were arranged as:

Normal_1 → Normal_2 → Normal_3 → TNB_1 → TNB_2 → TNB_3

This ordering places the Normal samples together and the TNB samples together, facilitating visual comparison between the two groups.

Figure 3. Heatmap of 94 differentially expressed genes across TNB and Normal samples.



The heatmap provides a visual representation of the relative expression patterns of the identified DEGs across the six samples.

5.8 Top Differentially Expressed Genes

The most prominent differentially expressed genes were further examined using the top DEG results.

The analysis generated:

top_20_DEGs.csv

This output provides a focused view of the genes showing the strongest differential expression according to the reported statistical measures.

The complete DEG results were retained alongside the top-gene subset to allow both detailed and summarized examination of the differential expression findings.

5.9 Gene Ontology Enrichment Analysis

Gene Ontology (GO) enrichment analysis was performed to investigate the biological functions associated with the identified differentially expressed genes. The upregulated and downregulated gene sets were analyzed separately to distinguish the functional characteristics associated with each direction of expression change.

5.9.1 Upregulated Genes

The 38 upregulated genes were subjected to GO enrichment analysis. The resulting enrichment output was retained as:

UP_GO_exploratory.csv

The analysis was used to identify enriched Gene Ontology terms associated with the upregulated gene set.

[INSERT UP_GO_exploratory.csv HERE]

The enriched GO terms provide an exploratory overview of the biological functions represented among the genes showing increased expression in TNB relative to Normal samples.

5.9.2 Downregulated Genes

The 56 downregulated genes were independently subjected to GO enrichment analysis.

The resulting output was retained as:

DOWN_GO_exploratory.csv

This analysis provides an exploratory overview of the biological functions represented among genes showing decreased expression in TNB relative to Normal samples.

5.9.3 Combined DEG GO Analysis

A combined GO analysis was also retained for the complete DEG set:

ALL_DEGs_GO.csv

The combined analysis provides an overall functional view of the 94 identified DEGs.

5.10 KEGG Pathway Enrichment Analysis

KEGG pathway enrichment analysis was performed separately for the upregulated and downregulated DEG sets.

The purpose of this analysis was to identify biological pathways represented by the differentially expressed genes and to provide additional functional interpretation of the transcriptional differences observed between TNB and Normal samples.

5.10.1 KEGG Enrichment of Upregulated Genes

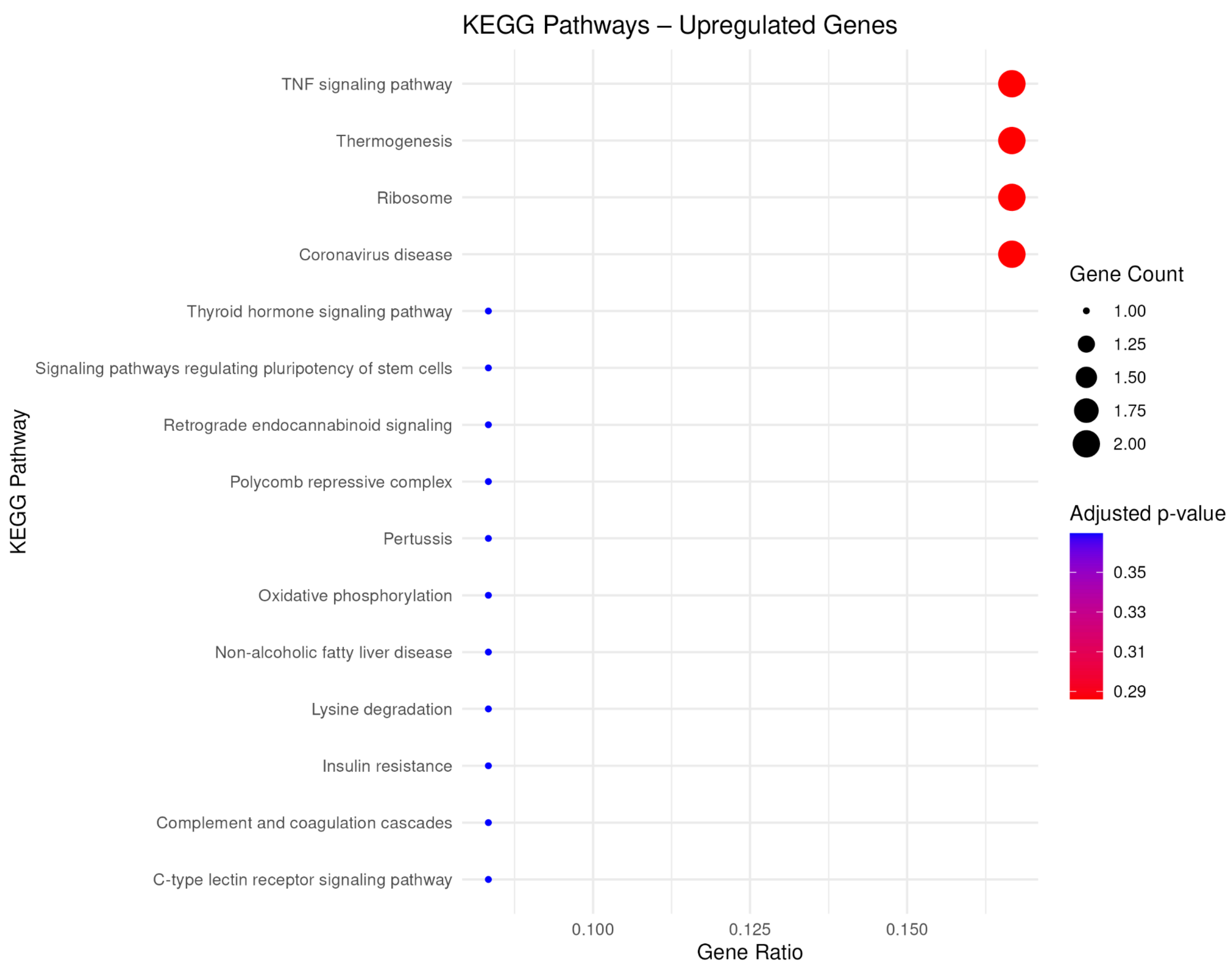
The 38 upregulated genes were analyzed for KEGG pathway enrichment.

The resulting pathway table was retained as:

UP_KEGG_pathways.csv

The corresponding dot plot was generated to visualize the pathway enrichment results.

Figure 4. KEGG pathway enrichment of upregulated genes.



Among the pathways showing the lowest nominal p-values were:

- TNF signaling pathway
- Ribosome
- Thermogenesis
- Coronavirus disease
- Lysine degradation

The TNF signaling pathway showed the lowest nominal p-value among the reported upregulated pathways.

However, the adjusted p-values of the leading pathways were above the conventional 0.05 threshold. Therefore, these findings are interpreted as **exploratory pathway enrichment rather than statistically significant pathway-level enrichment**.

5.10.2 KEGG Enrichment of Downregulated Genes

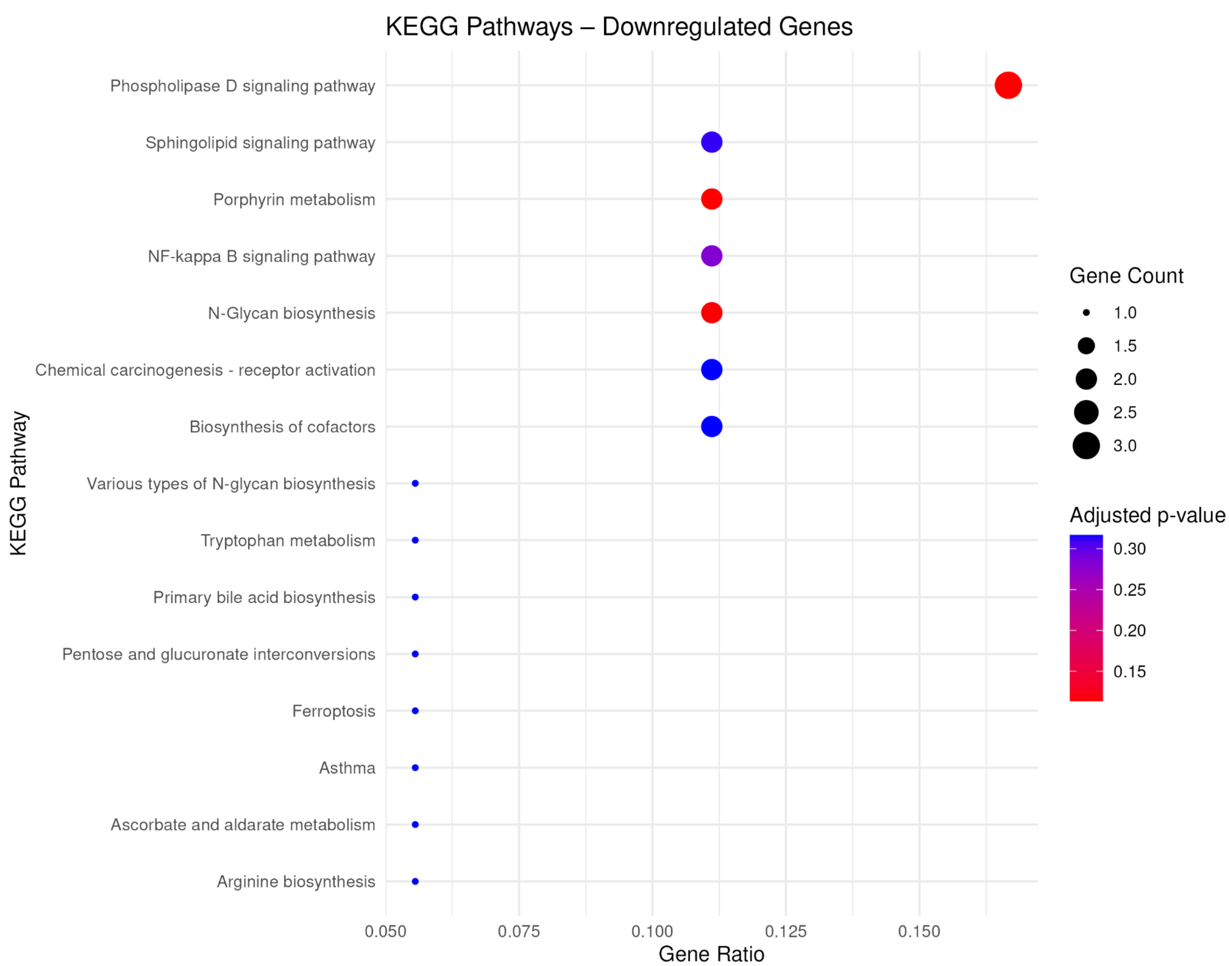
The 56 downregulated genes were independently analyzed for KEGG pathway enrichment.

The resulting pathway table was retained as:

DOWN_KEGG_pathways.csv

The corresponding enrichment visualization was generated as a dot plot.

Figure 5. KEGG pathway enrichment of downregulated genes.



The pathways with the lowest nominal p-values included:

- Phospholipase D signaling pathway
- Porphyrin metabolism
- N-Glycan biosynthesis
- NF-kappa B signaling pathway
- Sphingolipid signaling pathway

- Primary bile acid biosynthesis

The phospholipase D signaling pathway showed the lowest nominal p-value among the reported downregulated pathways.

As with the upregulated gene analysis, the adjusted p-values did not meet the conventional 0.05 significance threshold for the leading pathways. Therefore, these results should be regarded as **exploratory observations rather than definitive pathway-level findings**.

5.11 Downstream R Analysis

Following the Galaxy-based analysis, selected DEG outputs were imported into R for additional downstream processing.

The Ensembl gene identifiers were processed to remove version suffixes and converted to **Entrez Gene identifiers** using the org.Hs.eg.db annotation database.

Of the identified genes:

- **23 upregulated genes** were successfully mapped to Entrez identifiers.
- **47 downregulated genes** were successfully mapped to Entrez identifiers.

These mapped identifiers were subsequently used for KEGG enrichment analysis.

The R workflow also generated the final visualizations used in this report, including the DEG heatmap and volcano plot. The generated figures were uploaded back to the Galaxy History to maintain a centralized record of the analysis outputs.

6. DISCUSSION

6.1 Differential Gene Expression

The TNB versus Normal RNA-seq comparison identified **94 differentially expressed genes**, comprising **38 upregulated genes and 56 downregulated genes**. This indicates that the two sample groups exhibit measurable differences in their transcriptional profiles.

The separation of the DEG set into upregulated and downregulated genes was important for subsequent functional analysis because genes changing in opposite directions may contribute to different biological processes and pathways.

The volcano plot provides a graphical representation of these differential expression results, while the DEG tables provide the corresponding statistical measurements for individual genes.

6.2 Expression Patterns and Sample-Level Variation

PCA was used as an exploratory analysis to examine the overall structure of the expression data. By reducing the multidimensional expression dataset into principal components, PCA provides a visual representation of similarities and differences among the six samples.

The PCA analysis was complemented by the 94-DEG heatmap. The heatmap demonstrated the relative expression patterns of the identified DEGs across the Normal and TNB samples.

Together, PCA and heatmap visualization provide complementary information: PCA focuses on overall sample-level variation, whereas the heatmap allows individual DEG expression patterns to be examined across samples.

6.3 Biological Interpretation of Upregulated Genes

The 38 upregulated genes were analyzed separately using GO and KEGG enrichment approaches.

The KEGG analysis produced several pathways with relatively low nominal p-values, including the **TNF signaling pathway, Ribosome, Thermogenesis, and Coronavirus disease** pathways.

However, the adjusted p-values for the leading pathways were above the conventional statistical significance threshold. Therefore, these pathways cannot be described as statistically significant enrichment based on the adjusted results.

Instead, they represent **candidate pathways for further investigation** and provide exploratory information about biological functions represented within the upregulated DEG set.

6.4 Biological Interpretation of Downregulated Genes

The 56 downregulated genes were analyzed independently using GO and KEGG enrichment.

The leading nominally enriched KEGG pathways included **Phospholipase D signaling, Porphyrin metabolism, N-Glycan biosynthesis, NF-kappa B signaling, and Sphingolipid signaling**.

As with the upregulated gene set, the adjusted p-values for these pathways did not reach the conventional 0.05 significance threshold. Consequently, the results should be interpreted as exploratory pathway associations rather than definitive evidence of pathway activation or suppression.

The separation of the upregulated and downregulated gene sets nevertheless provides a useful framework for identifying biological themes that may warrant additional investigation.

6.5 Integration of Galaxy and R Analysis

A major strength of the project was the combination of **Galaxy-based workflow execution** with **R-based downstream analysis**.

Galaxy provided a reproducible environment for the major RNA-seq processing steps, including quality assessment, preprocessing, alignment, quantification, and differential expression analysis. The Galaxy History preserves the datasets and computational steps associated with the analysis.

R was subsequently used to perform additional processing, gene identifier conversion, enrichment analysis, and visualization. This allowed the final results to be explored in greater detail while maintaining the generated outputs within the Galaxy project.

This combined workflow demonstrates how graphical bioinformatics platforms and programmable statistical environments can be used together in a reproducible RNA-seq analysis.

6.6 Limitations and Interpretation

Several limitations should be considered when interpreting the results.

First, the analysis included **three biological replicates per condition**, which provides a basis for statistical analysis but represents a relatively limited sample size.

Second, although several KEGG pathways showed low nominal p-values, the leading pathways did not meet the conventional **adjusted p-value < 0.05** threshold. Therefore, pathway enrichment results should not be interpreted as definitive biological mechanisms.

Third, the identification of differentially expressed genes and pathway associations does not by itself establish biological causation. Experimental validation would be required to confirm the biological significance of individual candidate genes or pathways.

Therefore, the current analysis should primarily be considered a **transcriptomic discovery and exploratory analysis**, generating candidate genes and biological pathways for further investigation.

6.7 Overall Interpretation

Overall, the analysis demonstrates clear transcriptional differences between the TNB and Normal groups, with **94 DEGs identified from the differential expression analysis**. The combination of differential expression statistics, PCA, heatmap visualization, GO analysis, and KEGG analysis provides multiple complementary perspectives on these differences.

While the pathway enrichment results should be interpreted cautiously because of the adjusted significance results, they provide useful exploratory candidates for understanding the biological processes represented within the DEG sets.

The complete Galaxy History and associated output datasets provide a reproducible computational record of the analysis.

7. CONCLUSION

This project established a complete and reproducible RNA-seq analysis workflow for comparing **TNB and Normal samples**. The analysis was performed primarily using the Galaxy platform, followed by downstream processing and visualization in R and Bioconductor.

The workflow included sequencing quality assessment, read preprocessing, reference genome alignment, gene-level quantification, and differential expression analysis using DESeq2. The analysis identified **94 differentially expressed genes**, consisting of **38 upregulated genes and 56 downregulated genes** in the TNB versus Normal comparison.

PCA, volcano plot, and heatmap visualization were used to examine the overall expression patterns and relationships among the samples and identified DEGs. Subsequent GO and KEGG enrichment analyses provided functional and pathway-level characterization of the identified genes.

Although several KEGG pathways showed relatively low nominal p-values, the leading pathway results did not meet the conventional adjusted significance threshold. Therefore, these pathway findings were considered **exploratory** rather than definitive evidence of pathway activation or suppression.

Overall, the project demonstrates an end-to-end RNA-seq differential expression workflow combining **Galaxy-based bioinformatics processing with R-based statistical analysis and visualization**. The resulting DEG lists, expression visualizations, enrichment analyses, and preserved Galaxy History provide a reproducible foundation for further investigation of the transcriptional differences between TNB and Normal samples.

8. REFERENCES

Use the following references for the final report:

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11. **Galaxy Project.** Galaxy: an open, web-based platform for accessible, reproducible computational biomedical research.

9: Appendix

Appendix A — Galaxy History

Include the complete Galaxy History link again:

MP_403 — TNB vs Normal RNA-seq Galaxy History

Appendix B — Major Output Files

38_upregulated_DEGs.csv

56_downregulated_DEGs.csv

all_94_DEGs.csv

94_DEG_counts.csv

94_DEG_heatmap.png

94_DEG_heatmap_zscore.tsv

94_DEG_volcano_plot.png

PCA_coordinates.csv

PCA_DEGs_TNB_vs_Normal.png

top_20_DEGs.csv

UP_GO_exploratory.csv

DOWN_GO_exploratory.csv

UP_KEGG_pathways.csv

DOWN_KEGG_pathways.csv

UP_KEGG_dotplot.png

DOWN_KEGG_dotplot.png

Appendix C — R Downstream Analysis Was used to

- Ensembl ID processing
- Ensembl → Entrez conversion
- KEGG enrichment
- heatmap generation
- volcano plot generation
- downstream visualization